Reverse Engineering Preprocessor REPr

What is Software Reverse Engineering (RE)?

RE is the process of identifying the purpose of compiled code. It is often used by vulnerability researchers and malware analysts. RE can provide deeper insights into code structure and potential vulnerabilities.

The RE Process

- A physical device is acquired
- Firmware is extracted
- Firmware is processed into executable code
- Executables are decompiled
- Pseudo-source code is generated

Abstract

Processing extracted firmware is the most fraught step of the RE process; it is often difficult to identify the file format and compression schemes used. REPr is a modular file identification and extraction utility. Users can create plugins which add functionality, allowing more file types to be extracted or identified. Reverse engineers can save time by reducing duplicated effort and quickly moving from extracting firmware to decompiling it.

Without REPr

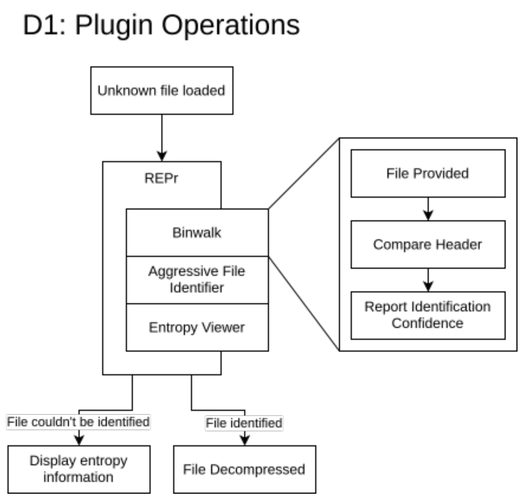
Executable code is time-consuming to extract from firmware. Teams often independently write their own extraction pipelines.

With REPr

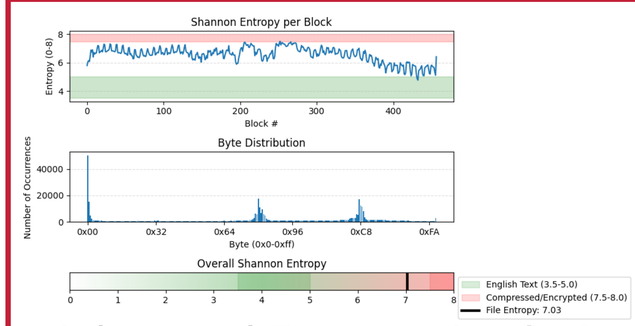
Firmware extraction is made mostly autonomous. Teams are able to share plugins, only solving problems once.

Function

REPr is a command-line utility which ingests a file and attempts to identify it with all loaded plugins. Each plugin returns a confidence score which is then used to determine how to extract contents from the file. In the event the file cannot be identified, other plugins can provide crucial information like entropy values to help a reverse engineer pick their next steps.



Features



Advanced Entropy Analysis

Recursive File Identification and Extraction

Quickly create and share bespoke processors...

...or choose from existing ones!

Modular and Highly Extensible

Hacking Competition Utilities

Jacob Greenberg
Advised by: Gary S.
Department: CEAS/CS

CEAS '26 EXPO